

WEEK 1 GENETIC ALGORITHM FOR OPTIMIZATION PROBLEM

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import random

employees = ["A", "B", "C", "D"]
days = ["Mon", "Tue", "Wed", "Thu", "Fri"]

POP_SIZE = 10
GENERATIONS = 50
MUTATION_RATE = 0.1

# Create a random schedule
def create_schedule():
    return [random.choice(employees) for _ in days]

# Fitness: avoid assigning the same employee
# too many times
def fitness(schedule):
    counts = [schedule.count(e) for e in employees]
    return 100 - max(counts) * 10

# Crossover
def crossover(p1, p2):
    point = random.randint(1, len(days) - 1)
    return p1[:point] + p2[point:]

# Mutation
def mutate(schedule):
    if random.random() < MUTATION_RATE:
        i = random.randint(0, len(days) - 1)
        schedule[i] = random.choice(employees)
    return schedule

# Initial population
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# Initial population
population = [
    create_schedule()
    for _ in range(POP_SIZE)
]

# Genetic Algorithm
for generation in range(GENERATIONS):

    # Selection
    population.sort(
        key=fitness,
        reverse=True
    )

    parents = population[:5]

    # Create new population
    new_population = parents.copy()

    while len(new_population) < POP_SIZE:

        p1 = random.choice(parents)
        p2 = random.choice(parents)

        child = crossover(p1, p2)
        child = mutate(child)

        new_population.append(child)

    population = new_population

# Best solution
best = max(population, key=fitness)

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    parents = population[:5]

    # Create new population
    new_population = parents.copy()

    while len(new_population) < POP_SIZE:

        p1 = random.choice(parents)
        p2 = random.choice(parents)

        child = crossover(p1, p2)
        child = mutate(child)

        new_population.append(child)

    population = new_population

# Best solution
best = max(population, key=fitness)

print("Best Schedule:")

for day, employee in zip(days, best):
    print(day, "->", employee)

print("Fitness:", fitness(best))

```

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... Best Schedule:
Mon -> D
Tue -> C
Wed -> A
Thu -> C
Fri -> D
Fitness: 80

```